Math 1610, Honors Calculus, Fall 2026, Worksheet 3

September 21, 2026

Name _____

Please write only your own name in the space above.

1. Recall that if f is defined on an interval around a, we say that $\lim_{x\to a} f(x) = L$ if for every $\epsilon > 0$ there exists a $\delta > 0$ such that
$$0 < |x - a| < \delta \implies |f(x) - L| < \epsilon.$$
Show that
$$\lim_{x\to a} f(x) = L \qquad \text{if and only if} \qquad \lim_{h\to 0} f(a + h) = L.$$